1 Classical statistical physics – Solutions generalities

The problems are roughly in order of difficulty. Those indicated with ♡ are those which it is most important to work on. Any labelled with ♣ are the hardest ones, which might only occur as a "sting in the tail" at the end of a long examination question.

♡*Problem 1.1* **Density of states for a free particle in one dimension**

(i) Note that nothing in the integrand depends on q.

$$\begin{aligned} W(E,L) &= \frac{1}{\mathsf{h}}\int_0^L \mathrm{d}q \int_{-\infty}^{\infty} \mathrm{d}p\ \delta\left(E - \frac{p^2}{2m}\right) \\ &= \frac{L}{\mathsf{h}}\int_{-\infty}^{\infty} \mathrm{d}p\ \delta\left(E - \frac{p^2}{2m}\right) \ . \end{aligned}$$

(ii) Since the integrand is even in p, we may replace $\int_{-\infty}^{\infty} \mathrm{d}p \cdots$ by $2\int_0^{\infty} \mathrm{d}p \cdots$.

(iii) Since $\varepsilon = p^2/(2m)$,

$$p = \sqrt{2m\varepsilon} \quad \Rightarrow \quad \mathrm{d}p = \frac{1}{2}\sqrt{\frac{2m}{\varepsilon}}\mathrm{d}\varepsilon \ ,$$

picking the positive square root as $p \geq 0$ after the last part of the question. Then

$$W(E,L) = \frac{2L}{\mathsf{h}}\int_0^{\infty} \mathrm{d}\varepsilon\ \frac{1}{2}\sqrt{\frac{2m}{\varepsilon}}\delta(E-\varepsilon) \ .$$

(iv) Letting the delta function do its work, we find:

$$W(E,L) = \frac{2L}{\mathsf{h}}\frac{1}{2}\sqrt{\frac{2m}{E}} = \frac{1}{E}L\sqrt{\frac{2mE}{\mathsf{h}^2}} = \frac{1}{E}\frac{L}{\lambda(E)} \ .$$

Problem 1.2 **All the space in high dimensions is in the corners**
The areas are:

$$A_2(S_{\mathsf{in}}) = 2 < A_2(\bigcirc) = \pi < A_2(S_{\mathsf{ex}}) = 2^2 = 4 \ .$$

The volumes are

$$V_3(C_{\mathsf{in}}) = \frac{8}{3^{3/2}} < V_3(\mathsf{sphere})\frac{4}{3}\pi < V_3(C_{\mathsf{ex}}) = 2^3 = 8 \ .$$

For n dimensions we have for the the exscribed hypercube has volume $V_n(C_{\mathsf{ex}}) = 2^n$, the inscribed hypercube has vertices at

$$\mathbf{r}_{\mathsf{v}} = a_n \underbrace{(\pm 1, \pm 1, \cdots, \pm 1)}_{n\ \mathsf{terms}} \ , \quad \text{and} \quad \mathbf{r}_{\mathsf{v}} \cdot \mathbf{r}_{\mathsf{v}} \ .$$

Thus $a_n = 1/\sqrt{n}$, and the sides have length $2/\sqrt{n}$, and the volume is $V_n(C_{\text{in}}) = 2^n/n^{n/2}$.

$$\frac{2^n}{n^{n/2}} < \frac{1}{\sqrt{n\pi}} \left(\frac{2\pi e}{n}\right)^{n/2} < 2^n \ .$$

If we divide through by 2^n we find:

$$\frac{1}{n^{n/2}} < \frac{1}{\sqrt{n\pi}} \left(\frac{\pi e}{2n}\right)^{n/2} < 1 \ .$$

Notice how the lefthand and middle terms tend to zero very rapidly as $n \to \infty$, ie $\sim n^{-n/2}$. Thus almost all of the volume lies between the hypersphere and the exscribed hypercube - ie in the corners of the hypercube.

Problem 1.3 **Stirling's approximation – at last derived**
We may write $n!$ as an integral (see sections 2.6 and 4.10.2 Y1 Maths notes), the Gamma function:

$$n! = \Gamma(n+1) = \int_0^\infty \mathrm{d}x \ x^n \mathrm{e}^{-x} \ .$$

The integral is a special case of Laplace's method. In general it is approximating an integral of the form

$$I = \int \mathrm{d}x \ \exp\left(g(x)\right) \ ,$$

by finding the maxima of $g(x)$ and Taylor expanding around those points (in the exponent). If there is only one stationary point (the case for $\Gamma(n+1)$), at say $x = x^*$, where $g'(x^*) = 0$, then we find (see question text for logic):

$$I = \exp\left(g(x^*)\right) \sqrt{\frac{2\pi}{|g''(x^*)|}} \ ,$$

a depiction of the integrand of the Gamma function is in Fig. (1.1).

For the Gamma function, $g(x) = n \ln x - x$, so:

$$\begin{aligned} g'(x^*) &= \frac{n}{x^*} - 1 = 0 \\ \Rightarrow \quad x^* &= n \\ g''(x) &= -\frac{n}{x^2} \\ \Rightarrow \quad g''(x^*) &= -\frac{n}{x^{*2}} \\ &= -\frac{1}{n} \ . \end{aligned}$$

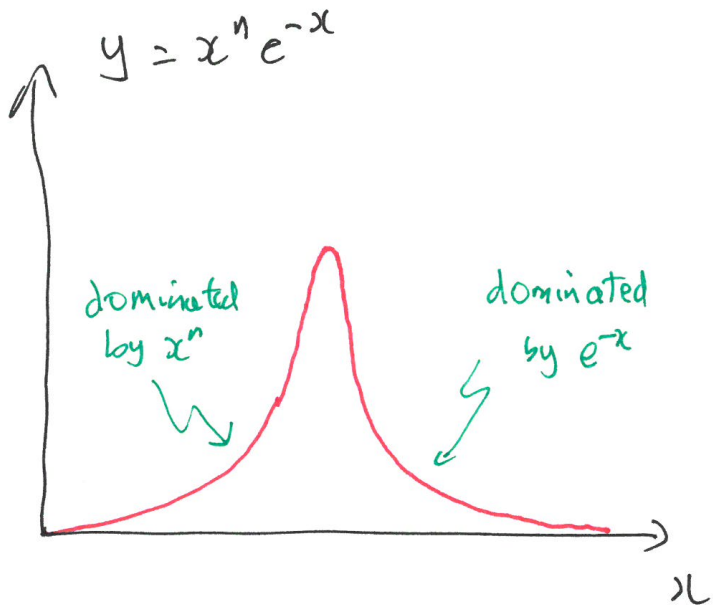

Figure 1.1: The integrand of the Gamma function - noting the maximum, which we approximate as a Gaussian function. As is shown the peak is centred at $x = x^* = n$ and the width is $\sim \sqrt{n}$, so the whole peak is well away from $x = 0$.

So

$$\begin{aligned} n! &= \Gamma(n+1) \\ &\underset{n\to\infty}{\sim} \exp\left(g(x^*)\right) \sqrt{\frac{2\pi}{|g''(x^*)|}} \\ &= \exp\left(n[\ln n - 1]\right) - \sqrt{\frac{2\pi}{|(1/n)|}} \\ &= n^n \mathrm{e}^{-n} \sqrt{2\pi n} \ . \end{aligned} \tag{1.1}$$

Problem 1.4 **Volume and surface area of a hypersphere**
By equating the two results for the Gaussian integral,

$$I_n = \int_{-\infty}^{\infty} \mathrm{d}x_1 \cdots \int_{-\infty}^{\infty} \mathrm{d}x_n \exp\left\{-a(x_1^2 + \cdots + x_n^2)\right\} = \int_0^{\infty} \mathrm{e}^{-ar^2} S_n r^{n-1} \, \mathrm{d}r$$

$$\Rightarrow \quad \left(\sqrt{\frac{\pi}{a}}\right)^n = S_n \int_0^{\infty} \mathrm{e}^{-ar^2} r^{n-1} \, \mathrm{d}r \ .$$

Now we change variables in the integral using

$$z = ar^2 \ , \Rightarrow \mathrm{d}r = \frac{1}{2\sqrt{a}} \frac{\mathrm{d}z}{\sqrt{z}} \ ,$$

to obtain

$$\begin{aligned}\int_0^\infty \mathrm{e}^{-ar^2} r^{n-1}\,\mathrm{d}r &= \frac{1}{2}\int_0^\infty \mathrm{d}z\ \mathrm{e}^{-z}\frac{1}{\sqrt{z}\sqrt{a}}\ \frac{z^{(n-1)/2}}{a^{(n-1)/2}}\\ &= \frac{1}{2}\frac{1}{a^{n/2}}\int_0^\infty \mathrm{d}z\ \mathrm{e}^{-z} z^{(n/2)-1}\\ &= \frac{1}{2}\frac{\Gamma\left(\frac{n}{2}\right)}{a^{n/2}}\ .\end{aligned}$$

We find upon substitution of this result

$$\begin{aligned}\left(\frac{\pi}{a}\right)^{n/2} &= \frac{1}{2}\frac{\Gamma\left(\frac{n}{2}\right)}{a^{n/2}} S_n\\ \Rightarrow\quad \frac{2\pi^{n/2}}{\Gamma\left(\frac{n}{2}\right)} &= S_n\ .\end{aligned} \tag{1.2}$$

Hence we may deduce V_n using $S_n = nV_n$ to obtain

$$V_n = \frac{S_n}{n} = \frac{2\pi^{n/2}}{n\Gamma\left(\frac{n}{2}\right)}\ ,$$

and, noting that

$$\frac{n}{2}\Gamma\left(\frac{n}{2}\right) = \Gamma\left(\frac{n}{2}+1\right)\ ,$$

We see

$$V_n = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2}+1\right)}\ . \tag{1.3}$$

We may now use Stirling's approximation at the level of $n! = n^n \mathrm{e}^{-n}\sqrt{2\pi n}$, to obtain:

$$\begin{aligned}V_n &\underset{n\to\infty}{\sim} \frac{\pi^{n/2}}{\left(\frac{n}{2}\right)^{n/2}\mathrm{e}^{-n/2}\sqrt{2\pi(n/2)}}\\ &= \frac{1}{\sqrt{\pi n}}\left(\frac{2\pi\mathrm{e}}{n}\right)^{n/2}\end{aligned}$$

For S_n, using Stirling's approximation we see

$$\Gamma\left(\frac{n}{2}\right) \underset{n\to\infty}{\sim} \left(\frac{n}{2}-1\right)^{n/2-1}\mathrm{e}^{-[n/2-1]}\sqrt{2\pi(n/2-1)}\ . \tag{1.4}$$

Let us simplify the third term in the product first:

$$\sqrt{2\pi(n/2-1)} = \sqrt{\pi n(1-2/n)} \simeq \sqrt{\pi n}(1-1/n) \simeq \sqrt{\pi n}\ . \tag{1.5}$$

We have decided to truncate at additive factors of order n^0.

Turning to the first two terms in Eq. (1.4), it is easier to compare magnitude of the logarithm:

$$\begin{aligned}
\left(\frac{n}{2}-1\right)^{n/2-1} \mathrm{e}^{-[n/2-1]} &= \exp\left[\left(\frac{n}{2}-1\right)\ln\left\{\frac{n}{2}\left(1-\frac{2}{n}\right)\right\}-\frac{n}{2}+1\right] \\
&\simeq \exp\left[\left(\frac{n}{2}-1\right)\left(\ln\frac{n}{2}-\frac{2}{n}\right)-\frac{n}{2}+1\right] \\
&= \exp\left[\frac{n}{2}\ln\frac{n}{2}-\frac{n}{2}-\ln\frac{n}{2}-1+1+\underbrace{\frac{2}{n}}_{\text{omit}}\right] \\
&= \left(\frac{n}{2}\right)^{n/2}\frac{2}{n}\,\mathrm{e}^{-n/2} \ .
\end{aligned}$$

Thus

$$\begin{aligned}
S_n = \frac{2\pi^{n/2}}{\Gamma\left(\frac{n}{2}\right)} &\underset{n\to\infty}{\sim} 2\frac{\pi^{n/2}}{\left(\frac{n}{2\mathrm{e}}\right)^{n/2}\frac{2}{n}\sqrt{\pi n}} \\
&= \left(\frac{2\pi\mathrm{e}}{n}\right)^{n/2}\sqrt{\frac{n}{\pi}} \\
&= nV_n \ ,
\end{aligned}$$

as it should.

Problem 1.5 **Most volume of a hypersphere is near its surface in high dimensions**

This is straightforward, we may take

$$V_n(R) = V_n(R=1)R^n \ .$$

Then consider a region $R > r > R - \mathrm{d}R$, The volume, ΔV_n is:

$$\begin{aligned}
\Delta V_n &= V_n(R=1)\left\{R^n-(R-\mathrm{d}R)^n\right\} \\
&\simeq V_n(R=1)\left\{R^n-R^n+n\,\mathrm{d}R\,R^{n-1}\right\} \\
&= V_n(R=1)n\,\mathrm{d}R\,R^{n-1} \ .
\end{aligned} \tag{1.6}$$

Consider the ratio, $\mathcal{V}$, of the volume of the shell, with thickness $\mathrm{d}R$, to the total volume of the hypersphere of radius R is:

$$\begin{aligned}
\mathcal{V} &= \frac{V_n(R=1)n\,\mathrm{d}R\,R^{n-1}}{V_n(R=1)R^n} \\
&= \frac{n\,\mathrm{d}R}{R} \ ,
\end{aligned}$$

this reflecting the enhancement of the importance of the surface. Of course the ratio may not be larger than unity, so $\mathrm{d}R$ must be smaller and smaller as $n \to \infty$ to justify the approximation in Eq. (1.6).

♡*Problem 1.6* **Spins, energy and temperature**
Consider N spins $\sigma_n = \pm 1$, $n = 1, 2, \cdots, N$ (they should be $\pm\frac{1}{2}$, but to make arithmetic simple ...). Each has a magnetic moment μ, and they all experience an external magnetic field $\mathbf{H} = H\hat{\mathbf{z}}$. The spins do not interact with each other, i.e. they are very dilute spatially.

(i) Write down the Hamiltonian $\mathcal{H}(\sigma_1, \sigma_2, \cdots, \sigma_N)$ in terms of $\{\sigma_n\}$, μ and H.

$$\mathcal{H}(\sigma_1, \sigma_2, \cdots, \sigma_N) = -\mu H \sum_{n=1}^{N} \sigma_n \ .$$

And then write the energy in terms of the "polarisation",

$$\frac{m}{\mu} = n_\uparrow - n_\downarrow \ .$$

The energy is

$$E = -\mu H(n_\uparrow - n_\downarrow) = -mH \ .$$

(ii) What are the maximum, E_{max}, and minimum, E_{min}, values of the energy for the spins in a field?

The maximum energy occurs when all spins are antiparallel to the field and the minimum is when they are all parallel. So:

$$E_{\text{max}} = -(-N\mu)H = N\mu H$$
$$E_{\text{min}} = -(N\mu)H = -N\mu H \ .$$

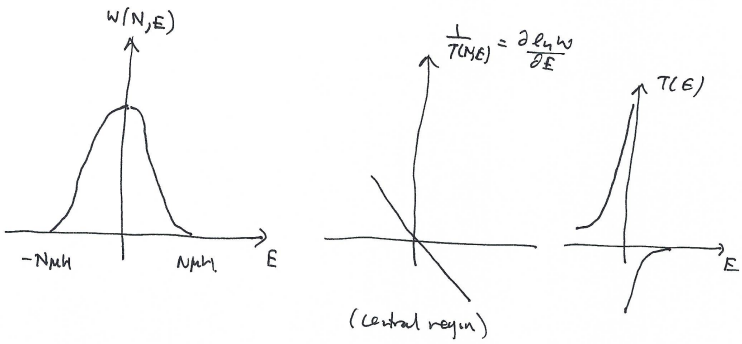

Figure 1.2: The number of configurations, $W(N, E)$, the reciprocal temperature and the temperature, $T(N, E)$.

(iii) Sketch the number of configurations, W, as a function of the polarisation, m/μ, $W(m/\mu)$ and hence as a function of E, $W(E)$.
See Fig. (1.2).

(iv) Determine the temperature using

$$k_{\text{B}} \frac{\mathrm{d} \ln W}{\mathrm{d}E} = \frac{\mathrm{d}S}{\mathrm{d}E} = \frac{1}{T(E)} \ .$$

and sketch $T(E)$ for $E_{\text{min}} \leq E \leq E_{\text{max}}$.[1]
See Fig. (1.2).

[1]We will see later that E is not the internal energy, $U(S, M)$, but actually the magnetic enthalpy, $\eta(S, H)$, and we may relate η to the temperature via $\mathrm{d}\eta = T\mathrm{d}S - M\mathrm{d}H$. So we are using $\partial\eta/\partial S|_H = T$ and identifying $E = \eta$.

(v) Is there something uncomfortable about the curve?

Yes, the temperature is infinite at $E = 0$, positively infinite as $E = 0$ is approached from negative values of E and negatively infinite as $E = 0$ is approached from $E > 0$.

(vi) Could such an eventuality arise with a gas of classical particles?

No, because there is no maximum energy. Or to be more precise there is is no maximum in $W(N, E)$ for a gas in free space - the number of states grows without limit, monotonically.

But a gas in a periodic potential is a different matter - as there can be a stationary points in the density of states.

Problem 1.7 **Two bodies in mechanical contact**

We have an overall enclosure of volume V_{tot}, with a mobile partition between the two sub-enclosures (forming the two bodies). The number of microstates $W_1(E_1, V_1)$ and $W_2(E_{\text{tot}} - E_1, V_{\text{tot}} - V_1)$, with

$$W_{\text{tot}} = W_1(E_1, V_1)W_2(E_{\text{tot}} - E_1, V_{\text{tot}} - V_1) \ .$$

We maximise this both with respect to E_1, as in lectures, and V_1. For E_1. We find in the former case, as before, that $\partial S/\partial E_1 = 1/T_1 = 1/T_2$.

Turning to the maximisation of the entropy with respect to V_1 we see:

$$\begin{aligned}
\frac{\partial W_{\text{tot}}}{\partial V_1} = 0 = \frac{\partial W_1(E_1, V_1)}{\partial V_1} W_2(E_{\text{tot}} - E_1, V_{\text{tot}} - V_1) + \\
+ W_1(E_1, V_1) \left. \frac{\partial \ln W_2(E_{\text{tot}} - E_1, V_2)}{\partial V_2} \right|_{V_2 = V_{\text{tot}} - V_1} \underbrace{\frac{\partial (V_{\text{tot}} - V_1)}{\partial V_1}}_{=-1}
\end{aligned}$$

$$\begin{aligned}
\Rightarrow \quad \frac{\dfrac{\partial W_1(E_1, V_1)}{\partial V_1}}{W_1(E_1, V_1)} &= \frac{\dfrac{\partial W_2(E_{\text{tot}} - E_1, V_2)}{\partial V_2}}{W_2(E_{\text{tot}} - E_1, V_2)} \\
\Rightarrow \quad \frac{\partial \ln W_1(E_1, V_1)}{\partial V_1} &= \frac{\partial \ln W_2(E_{\text{tot}} - E_1, V_2)}{\partial V_2} \\
\Rightarrow \quad \frac{p_1}{T_1} &= \frac{p_2}{T_2} \ ,
\end{aligned}$$

where we used the first law in form provided in the question:

$$\mathrm{d}S = \frac{1}{T}\mathrm{d}U + \frac{p}{T}\mathrm{d}V \ .$$

Note if we did not maximise *both* E and V, but only V, we could only deduce the ratios of p and T were equal, not the variables separately:

$$\left(\frac{p}{T}\right)_1 = \left(\frac{p}{T}\right)_2 \ .$$

Problem 1.8 **The isobaric ensemble**

The total number of configurations, with fixed energy and volume in both the system and the bath is

$$W_{\text{tot}}(E_{\text{s}}, E_{\text{tot}}; V_{\text{s}}, V_{\text{tot}}) = W_{\text{s}}(E_{\text{s}}, V_{\text{s}})W_{\text{b}}(E_{\text{tot}} - E_{\text{s}}, V_{\text{tot}} - V_{\text{s}}) \ .$$

So the total number of configurations, adding all the ways to partition the energy and volume between the bath and the system, is

$$W_{\text{tot}}(E_{\text{tot}}, V_{\text{tot}})) = \int \mathrm{d}E_{\text{s}} \int \mathrm{d}V_{\text{s}} W_{\text{s}}(E_{\text{s}}, V_{\text{s}}) W_{\text{b}}(E_{\text{tot}} - E_{\text{s}}, V_{\text{tot}} - V_{\text{s}}) \ .$$

Then we may Taylor expand the second term up in an exponential (as before to get Boltzmann factor), but now in terms of both E_{s} and V_{s}. Then

$$\begin{aligned}
W_{\text{tot}}(E_{\text{tot}}, V_{\text{tot}}) &= \int \mathrm{d}E_{\text{s}} \int \mathrm{d}V_{\text{s}}\ W_{\text{s}}(E_{\text{s}}, V_{\text{s}}) \exp\left(\ln W_{\text{b}}(E_{\text{tot}} - E_{\text{s}}, V_{\text{tot}} - V_{\text{s}})\right) \\
&\simeq \int \mathrm{d}E_{\text{s}} \int \mathrm{d}V_{\text{s}}\ W_{\text{s}}(E_{\text{s}}, V_{\text{s}}) W_{\text{b}}(E_{\text{tot}}, V_{\text{tot}}) \exp\left[-E_{\text{s}} \left.\frac{\partial \ln W_{\text{b}}}{\partial E_{\text{b}}}\right|_{E_{\text{tot}}} - V_{\text{s}} \left.\frac{\partial \ln W_{\text{b}}}{\partial V_{\text{b}}}\right|_{V_{\text{tot}}}\right] \\
&= \int \mathrm{d}E_{\text{s}} \int \mathrm{d}V_{\text{s}}\ W_{\text{s}}(E_{\text{s}}, V_{\text{b}}) W_{\text{b}}(E_{\text{tot}}, V_{\text{tot}}) \exp\left(-\frac{E_{\text{s}}}{k_{\text{B}}} \frac{\partial S_{\text{b}}}{\partial E_{\text{b}}} - \frac{V_{\text{s}}}{k_{\text{B}}} \frac{\partial S_{\text{b}}}{\partial V_{\text{b}}}\right) \\
&= \int \mathrm{d}E_{\text{s}} \int \mathrm{d}V_{\text{s}}\ W_{\text{s}}(E_{\text{s}}, V_{\text{s}}) W_{\text{b}}(E_{\text{tot}}, V_{\text{tot}}) \exp\left(-\frac{E_{\text{s}}}{k_{\text{B}} T_{\text{b}}} - \frac{p_{\text{b}} V_{\text{s}}}{k_{\text{B}} T_{\text{b}}}\right) \ ,
\end{aligned}$$

where T_{b} and p_{b} are the temperature and pressure of the bath. We now discard the bath labels, and use T and p.

Then we may divide by the number of bath microstates (as it is a constant with respect to the integration) to leave an expression only involving the system states:

$$Y(T, p, N) = \frac{W_{\text{tot}}(E_{\text{tot}}, V_{\text{tot}})}{W_{\text{b}}(E_{\text{tot}}, V_{\text{tot}})} = \int \mathrm{d}E_{\text{s}} \int \mathrm{d}V_{\text{s}}\ W_{\text{s}}(E_{\text{s}}, V_{\text{s}}) \exp\left(-\frac{E_{\text{s}}}{k_{\text{B}} T} - \frac{p V_{\text{s}}}{k_{\text{B}} T}\right) \ . \quad (1.7)$$

Problem 1.9 **Relation to the Gibbs free energy**
Writing

$$W_{\text{s}}(E_{\text{s}}, V_{\text{s}}) = \exp\left(\frac{S_{\text{s}}(E_{\text{s}}, V_{\text{s}})}{k_{\text{B}}}\right) \ ,$$

we see that we may write the expression for the partition function as:

$$\begin{aligned}
Y(T, p, N) &= \int \mathrm{d}E_{\text{s}} \int \mathrm{d}V_{\text{s}}\ \exp\left(\frac{S_{\text{s}}(E_{\text{s}}, V_{\text{s}})}{k_{\text{B}}} - \frac{E_{\text{s}}}{k_{\text{B}} T} - \frac{p V_{\text{s}}}{k_{\text{B}} T}\right) \\
\left(\text{define } \beta = \frac{1}{k_{\text{B}} T}.\right) \quad &= \int \mathrm{d}E_{\text{s}} \int \mathrm{d}V_{\text{s}}\ \exp\left(-\beta[E_{\text{s}} - T S_{\text{s}} + p V_{\text{s}}]\right) \ .
\end{aligned}$$

We immediately recognise in the exponential

$$G(T, p, N) = U - TS + pV \ ,$$

the Gibbs free energy which is minimised at fixed pressure, suggesting

$$G(T, p, N) = -k_{\text{B}} T \ln[Y(T, p, N)] \ .$$

We must then perform the same analysis as in lectures for the Helmholtz free energy

We note that there must be at least one maximum of the integrand in Eq. (1.7): W_{s} is an increasing function of E_{s} and V_{s}, and the exponential decreases with both those variables.

We find the stationary point of the exponent, remembering (apart from places where $f(x) = -\infty$):

$$\frac{\mathrm{d}}{\mathrm{d}x}\,\exp\left(f(x)\right) = f'(x)\exp\left(f(x)\right) = 0 \quad \Rightarrow \quad f'(x) = 0\ .$$

Thus:

$$\frac{\partial}{\partial E_{\mathsf{s}}}\left(E_{\mathsf{s}} - TS_{\mathsf{s}} + pV_{\mathsf{s}}\right) = 0$$
$$\Rightarrow \quad 1 - T\frac{\partial S_{\mathsf{s}}}{\partial E_{\mathsf{s}}} = 0\ .$$

Then the value of the system (internal) energy, $E_{\mathsf{s}} = U$, at the stationary point is defined

$$\frac{1}{T} = \left.\frac{\partial S_{\mathsf{s}}}{\partial E_{\mathsf{s}}}\right|_{E_{\mathsf{s}}=U}\ .$$

Similarly for the volume

$$\frac{\partial}{\partial V_{\mathsf{s}}}\left(E_{\mathsf{s}} - TS_{\mathsf{s}} + pV_{\mathsf{s}}\right) = 0$$
$$\Rightarrow \quad -T\frac{\partial S_{\mathsf{s}}}{\partial V_{\mathsf{s}}} + p = 0\ .$$

Thus the volume, V^*, is determined by ensuring the pressure, from the derivative of the entropy, is consistent with the applied pressure, i.e.

$$-T\frac{\partial S_{\mathsf{s}}}{\partial V_{\mathsf{s}}}(V^*, T, N) = -p$$

is an implicit equation for V^*.

At this stationary point the corrections are then at second order - as in the case of the Helmholtz free energy.